Towards Joking, Humor Sense Equipped and Emotion Aware Conversational Systems

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Abstract. In this paper, we present our progress so far in realization of project aimed to create a complex, modular humor-equipped conversational system. By complex, we mean that it should be able to: 1) detect users' emotions, 2) detect users' humorous behaviors and react to them properly, 3) generate humor according to users' emotive states and 4) learn each user's individual sense of humor. The research is conducted in Japanese. We chose puns as a relatively computable genre of humor.
We describe a general outline of our system, as well as its four modules: humor detection module, emotion recognition module, response generator module and individualisation module. We present the algorithm of systems used in each module, along with some evaluation results.

Keywords: humor, puns, emotions, artificial intelligence, natural language processing, human-computer interaction

1 Introduction

In recent years the world of science has begun to realize that designing systems able to interact with humans in a natural manner requires considerations of such human factors as humor or emotions. These are especially important in conversational systems that interact with us on daily basis – e.g. virtual companions or interactive car navigators. To be perceived as friendly, natural and human-like, such systems need to be capable of detecting users' emotive states and react to them properly. Also, as humans sometimes tend to react to particular emotions with humor, these emotion-aware systems should also be able to detects and perform humorous behaviors.

That said, these reactions and humorous behaviors are highly individual and subjective issues, i.e. they strongly depend on particular human's personal traits, such as emotional intelligence and sense of humor. Thus, when constructing emotion-aware humor-equipped systems it is crucial to implement also an individualization procedure, that would allow the system to learn each particular user's humor sense and use this information to individualize its performance.

To summarize, the goal of this research is to create a conversational system able to: 1) detect users' emotions, 2) detect users' humorous behaviors and react to them properly, 3) generate humor according to users' emotive states and 4) learn each user's individual sense of humor. As humor is quite a broad issue, we decided to focus on its linguistic genre, known as verbal humor (or puns), which is said to be relatively easy to process (with Natural Language Processing tools). Puns are generally present in almost every culture, but in highly homophonic languages there are more possibilities of their creation. Therefore, we decided to conduct our research in one such language, namely Japanese.

This paper describes the progress of our project so far. First, we summarize the outline of our project. Next, we describe the system's modules: humor detecting module, emotion recognition module, response generating module and individualisation module, at their current stage of development (along with some evaluation experiments' results).

2 State of the art

In this section, we briefly present state of the art in the fields of humor detection, emotion recognition and humor generation.

2.1 Humor detection

As far as state of the art in the field of humor detection is concerned, here we focus on works that deal with detecting Japanese puns (*dajare*), as they are quite characteristic and thus require individual approach. A system classifying *dajare* was proposed Yokogawa [1]. Basing on results of morphological analysis, the system was able to successfully detect punning sentences – however, only those found as ungrammatical. Yokogawa also focused on imperfect puns, while perfect (full homophony-based) puns are also an important group.

Amaya et al. [2] proposed a method of automatic detection of word similarities based narrative jokes extracted from Twitter entries. To do so, they used WordNet and methods such as bag-of-words and machine learning.

Also worth mentioning is the work of Kitagaki [3], who proposed a fuzzy-logic based method of measuring *dajare* funniness (laughableness). This approach, however, a priori assumes that inputted texts are puns, and the author does not propose any method of distinguishing them from non-humorous texts.

2.2 Emotion recognition

Text based affect analysis is defined as a field focused on developing natural language

processing techniques for estimating the emotive aspect of text [4]. Research in this field has flourished rapidly through several years. Elliott [5], for example, proposed a keyword-based system which used an affect lexicon (including words like “happy”, or “sad”) along with intensity modifiers (like “extremely”, “somewhat”). Liu et al. [6] proposed a model of text-based affect detection based on a generic common sense database. Alm et al. [7] applied machine learning methods for affect analysis, which they tested on fairy tales. Machine learning was also used by Aman and Szpakowicz to perform affect analysis of blog entries [8].

Some work on affect analysis was also done for the Japanese language. Tsuchiya et al. [9] proposed a system using association mechanism to estimate emotive aspect of utterances. There were also attempts of using the Internet as a source of potentially emotive sentence examples [10, 11]. One such Web-based technique was also used by Ptaszynski et al. [12] as a support for his affect analysis system for Japanese text-based utterances.

2.3 Humorous responses to emotional states

In the last decades, numerous research projects were conducted on humor generation. In this section we focus on verbal humor and mostly on those works in which humor generation was placed in interactions with human users.

One of the first and most notable attempts at creating a computer system able to tell puns was Binsted’s JAPE punning riddles generator [13]. It was a stand-alone application, i.e. it generated riddles (such as: “-What do you call a murderer with fibre? -A cereal killer”) in isolated forms. However, JAPE was later used by Ritchie et al. [14] to create STANDUP – a system described as “a language playground, with which a child can explore sounds and meanings by making up jokes, with computer assistance. This gave the riddles environment and context, in which they are used naturally by humans.

Also worth noting is Tinholt and Nijholt’s ConceptNet-based cross-reference joke generator [15], implemented into a conversational system. Toward users’ utterances the system, when possible, would generate humorous misunderstanding including responses, using syntactic ambiguities (like: “User: Did you know that the cops arrested the demonstrators because they were violent?, System: The cops were violent? Or the demonstrators?”). Unfortunately, experiments showed that this type of cross-reference ambiguity occurs very rarely in real-life conversations, and thus the use of such generator in daily dialogues with conversational systems would be marginal. However, it would be interesting to investigate this issue also for languages other than English – Japanese, for instance, is a highly contextual language, with very limited range of grammar forms, and thus more potential syntactic ambiguities.

Another important work in the field of humor generation during conversations with users was done by Sjoebergh and Araki [16]. Their Japanese chatterbot used humor modules: database jokes module (from preprepared DB), similar dirty words module (generates dirty misunderstanding jokes, based on word similarities) and user jokes module (checks if user’s utterances can be found in jokes DB). This system received positive reactions from users.

However, to the authors’ best knowledge, no system able to react to users’ particular emotional states with humor has been created so far. Also, no humor-

equipped chatterbot has been built that would include individualization module, i.e. that would be able to adapt to users' humor sense. Thus, here lies the originality of our project – the system we are working on not only represents what we defined as multi-stage approach to humor and emotions [17], but also will probably be the first system able to model users' individual humor preferences.

3 System proposal

As mentioned above, some interesting work has been done in the fields of humor generation and recognition. There were also attempts at placing such systems in conversations with human users.

However, when constructing a system able to use humor in natural manner in daily interactions with users, we must ask ourselves a question: what should such system be able to do?

This would probably trigger a long lasting debate, but if we consider traits humans use when dealing with humor, we can state, that to do so, such system should at least be able to:

- detect user's humorous behaviors
- react to them adequately
- recognize users' emotions
- on this basis - decide whether or not use humor
- generate humorous contents
- recognize users' reactions to humorous contents
- construct users' humor sense models to individualize its performance.

Needless to say, humor is a quite complex process and these requirements may seem general and not sufficient. However, as to our knowledge no system exists that would fulfil them, we decided to launch a research project aimed at creating one.

Thus, the abovementioned requirements define our main research goal. Currently we are working on implementation and improvement of particular modules of the final system. Its general algorithm outline is presented on Figure 1.

The system is modular, i.e. it is constructed from several modules: humor detection module (see 4.1), emotion recognition module (4.2), response generation module (with three submodules: humorous response generator, non-humorous response generator and reaction to humor generator – see 4.3) and individualization module (4.4).

As shown in Figure 1, the algorithm works as follows: during a conversation between a user and the system, user's utterance is first analyzed by the humor detection module. If it recognizes the utterance as humorous, the response generation module generates an appropriate response. If the utterance is recognized as non-humorous, next it is analyzed by the emotion recognition module. Based on this analysis results, the system decides if humor should be used to enhance the user's mood. If yes, the sys tem's response is generated by the humorous response generator. If no, response is generated by the non-humorous response generator (i.e. baseline, non-humor- equipped chatterbot).

All these modules are linked to the individualization module, which, using the results of affect analysis, gathers data needed to build the user's individual humor sense module. If, for example, the user after being told a joke reacts to it with anger, the individualization module stores this data to be used in further interactions. The more such data the system gathers, the more complex model it is able to build. Complex individual models of users' sense of humor, in turn, will result in better and more personalized interactions.

More detailed modules description is given in section 4.

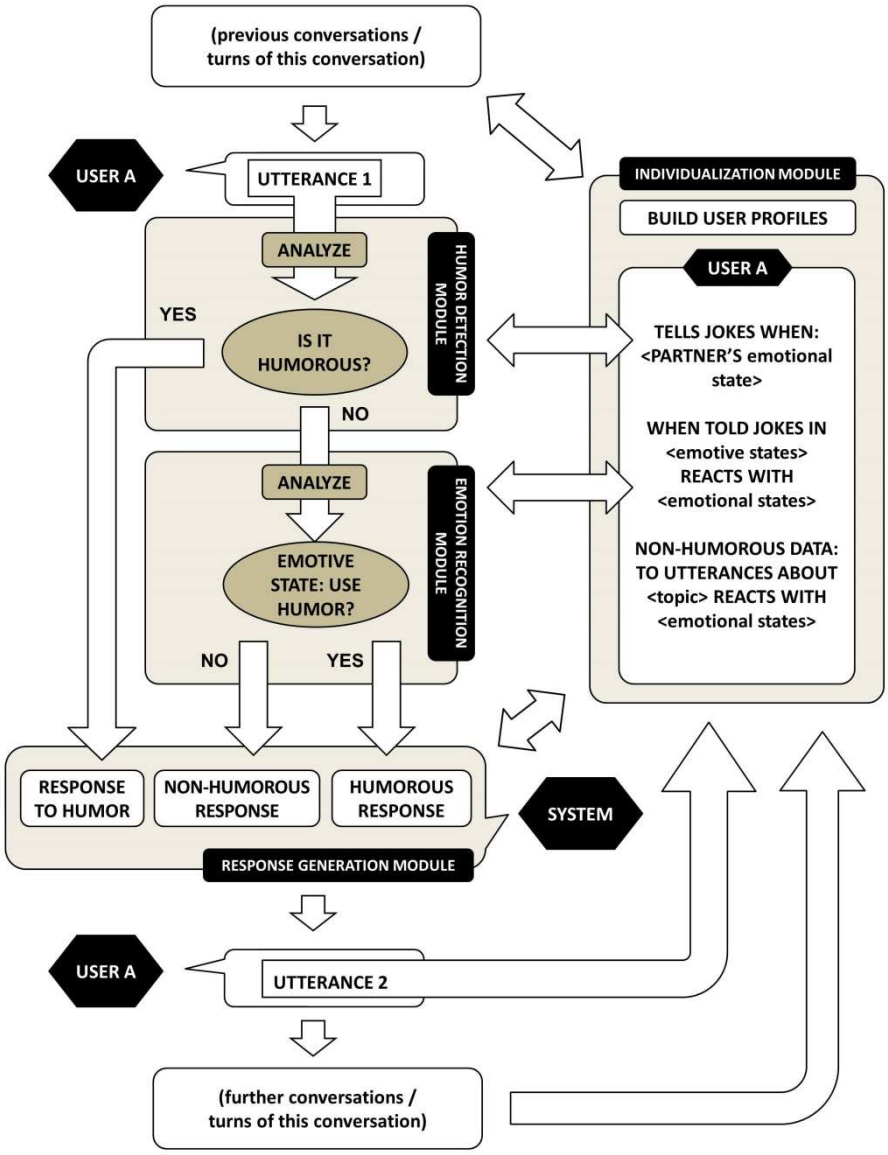


Fig. 1. Proposed system's algorithm outline

4 Modules

In this section we describe the modules of our system. Currently we are focusing on improving the humor detection module's performance and implementation of the

individualization module's algorithm. We are also working on adding new humor generation patterns to extend the humorous response generator's possibilities.

4.1 Humor detection module

We prepared a pun/non-pun classifier using an Support Vector Machine [18] with the features listed in Table 1. The positive examples were all obtained from existing web-sites that collect Japanese puns. The negative examples were sentences with limitation of sentence length, which was in order not to have too long sentences classified as false positive, as such sentences are more likely to be detected as puns.

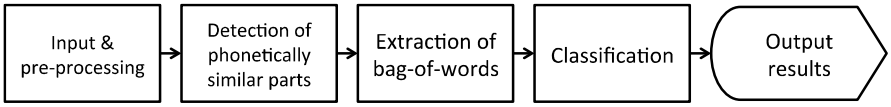


Fig. 2. Process flowchart of pun detection module.

The feature sets include phonetic similarity feature in addition to bag-of-words and phonetic exact-match feature. The phonetic exact-match feature checks whether there is a morpheme in the input sentence which sound is entirely equal to an additional part in the sentence. On the other hand, the phonetic similarity feature is related to how similar consonants in the input sentence are. It has value of 1 if the mean of the similarities for all combinations of consonants p1 and p2, which are calculated preliminary using development corpus data, is larger than the threshold that is given by pre-tuning. Formula (1) shows the method of this calculation. In this equation, Np1, p2 stands for the number of consonant pair p1 and p2 which occurs sequentially in a pun sentence, Npair for the number of all such pairs, P(p) for relative frequency of consonant p in all setnences, and c for a real number determined empirically.

$$\mathrm{OER}(p_1, p_2) = \log \frac{N_{p_1,p_2} + N_{p_2,p_1}}{N_{\mathrm{pair}} \cdot \mathrm{P}(p_1) \cdot \mathrm{P}(p_2)} + c \tag{1}$$

In order to verify performance of the pun detection module, we performed a 20-fold cross validation of the classifier. The size of both the examples were 23,984 sentences, which were divided into 22,784 versus 1,200 sentences for training and tested data, in which the same numbers of positive and negative examples exist.

Table 1. Feature sets used in the classifier performance measure experiment and its results

Set	BOW features	Exact-match	Similarity	Euphonic sounds	Precision	Recall	F-measure
F_1	-	O	O	O	0.908	0.861	0.884
F_2	O	-	-	-	0.873	0.824	0.848
F_3	O	O	O	O	0.915	0.633	0.749

The results of the experiment stated above are shown in **Table 1**. Using the feature set F_3, there was 0.14 point of improvement in F-value compared to the rule-based feature set (F_1). This suggests that it appears to have effectiveness to take vocabulary features into consideration in detecting sentences that include puns.

The results, albeit not perfect, show that this pun detection algorithm can be used in our system. Currently we are working on its full implementation, which will be evaluated experimentally in the nearest future.

4.2 Emotion recognition module

The main part of the emotion recognition module is ML-Ask system - a text-based affect analyzer for Japanese, developed by Ptaszynski et al. [19, 20].

ML-Ask is a keyword-based language-dependent system for automatic affect annotation on utterances in Japanese. It uses a two-step procedure:

1. Specifying whether an utterance is emotive - i.e. if it contains emotive elements (like “whoa!” or “Oh!”), which indicate that a sentence is emotive, but do not detail what specific emotions have been expressed, and
2. Recognizing the particular emotion types in utterances described as emotive – by detecting if it contains emotive expressions, i.e. words or phrases that define specific emotional states (like “happy” or “sad”).

To perform these two steps of analysis, ML-Ask uses a lexicon of emotive elements (created by Ptaszynski et al. [19], as well as a database of emotive expressions. These expressions in ML-Ask were also mapped on Russell’s two dimensional affect space [21], which assumes that all emotions can be represented in two dimensions: their valence (positive/negative) and activation (activated/deactivated).

Another important feature of ML-Ask is the use of Shi et al.’s web mining technique [10] to extract emotion associations from the Internet. This algorithm uses noun-plus-verb and noun-plus-adjective phrases found in inputted utterance to extract emotive associations, i.e. emotive expressions that co-occur with these phrases most frequently. For example, if for the input “I lost my job” the system would extract associations like “sad” or “angry”, as these are the groups of emotions most frequently used after this phrase.

ML-Ask is also able to detect specified emotions from emoticons (see [22] for details). This should be particularly useful in detection of users’ humorous utterances, as people generally tend to mark funny contents with emoticons.

The ML-Ask system is currently under further development. Particular focus is laid on automatic construction of an emotive expressions dictionary, based on sentence

annotations performed by humans. Also, Ptaszynski et al.'s SPEC system [....] will be used to analyze patterns that occur in emotive sentences.

The ML-Ask system plays also a pivotal role in the individualisation mode (see 4.4), in which it is used to analyze users' emotive states during conversations. On this basis the system builds each user's personal model of preferences.

4.3 Response generation module

The system's response to user's utterance is generated by the response generation module. Depending on the individualization module's decision regarding the type of response, it is generated by one of three submodules: non-humorous response generator, humorous response generator and response to user's humor generator.

Non-humorous response generator (chatterbot). This submodule is responsible for generating responses, that are neither humorous nor reactions to humor. In other words, it functions as a standard chatterbot that performs conversations with users. In our previous works we used Modalin, developed by Higuchi et al. [24, 25] Modalin is a keyword-based chatterbot that uses the Internet to generate word associations to phrases extracted from users' utterances. Experiments showed that Modalin's utterances received fairly good evaluation from users; however, some of them were found not natural or ungrammatical. Also, as Modalin performs many queries on the Internet, its processing time is sometimes too long to use the system in real time conversations.

Thus, for the needs of this project, we decided to improve Modalin's performance. The previous version of the system used sentence modalities in a random manner, i.e. it added randomly selected modal phrases to sentences generated by the system. In the second version, we decided to create a dictionary containing modal expressions and grammar forms they tend to co-occur with. This allows the system to avoid generating unnatural and grammatically incorrect sentences. In addition, to avoid Internet lag issues, we switched the system to an offline text corpus.

Humorous response generator. As at this stage of our research we focus on verbal humor (puns), the system's humorous responses are generated by a pun generator. Its basic version, PUNDA system, was developed in our earlier works [25]. The system first extracts particular words (nouns, adjectives and verbs) from inputted sentence and then uses them as potential pun base phrases, i.e. it attempts to transform them into other phonetically similar words. Next, the phrases are transformed using phonetic patterns based on our Japanese puns classification [26]. Then, the system uses the Internet to detect which candidates are relevant to the input and uses sentence templates to generate humorous utterances. The system generates pun candidates with 75% accuracy (compared to human level). Below is an example of the system's process:

User: *Kaeru daikirai*! (I hate frogs!)
Base word: *kaeru* (frog)
Pun candidate: *tsukaeru* (be able to use)]
System's response: *Kaeru to ieba tsukaeru ne*! (Speaking of frogs, we can use them!)

In our previous research [25, 26] PUNDA system was implemented into a conversational system. Experiments, in which human users were asked to perform conversations with two systems: with and without humor, showed that the former was evaluated as better, more human-like and triggering more positive emotions than the latter.

Currently we are working on improving this submodule's performance. This will be achieved by adding more pun generation patterns and sentence templates as well as refining the association generation algorithm.

Response to user's humor generator. When the humor detection module recognizes that user told a joke, the system's response is generated by the response to user's humor submodule.

The easiest method to make the system react to users' jokes is to make it return replies like "hahaha" or "that was funny!" – as it was done by Sjoebergh and Araki [16]. At this stage of our research, we present similar approach. We prepared 5 different phrases that can be used in reaction to users' humorous utterances.

However, needless to say, humans do not react to all types of humor in similar manner. There is high possibility that poor or old jokes may not be found particularly funny. Depending on topic, some jokes may be perceived as offensive or aggressive. This, on the other hand, is also highly individual, as what seems offensive to one person might be found funny by another.

Therefore, in the nearest future we are planning to modify the humor detection module and add to it what we call a humor evaluation submodule – a system that will be able to assess the quality of users' humor. To do so, we are going to have human evaluators assess existing examples of humorous utterances and use the statistical results in the system.

Another issue linked to the humor assessment is the system's sense of humor. In our research so far we have been focusing on the system's ability to personalize its performance acording to users' preferences. However, a question should be asked, whether such system should be able to develop its own original sense of humor, and to what extent should it influence its performance. In the future we are planning to investigate this issue.

4.4 Individualization module

The individualization module is responsible for the system's ability to adapt to users' personal preferences. It uses the ML-Ask affect analyzer (see 4.2) to gather data about users' emotive states during conversations. On this basis the module builds each user's individual model of sense of humor. The data acquisition is performed as showed in Table 2.

Table 2. Subjects of affect analysis and data acquired by the individualisation module

Subject of affect analysis	User data acquired
The system's utterance preceding user's humorous utterance	Emotive states to which the user tends to react with humor
User's utterance preceding the system's humorous utterance	- User's emotive reactions to the system's humorous utterances (humor genre, topic etc.) - User's emotive reaction to the system's humorous utterances generated toward user's particular emotive states
User's utterance following the system's humorous utterance	

If, for example, the emotiveness analysis module detects that User A's emotive state is <sadness>, the individualization module analyzes his/her humor sense model to check this user's reactions in similar situations that occurred in previous conversations. This entry of humor sense model may look as follows:

User A
emotion changes when told jokes in emotive state <sadness>:
- to positive: 7/10
- to neutral: 2/10
- to negative: 1/10

On this basis, the system decides that it should use humor (i.e. the response should be generated by the humorous response generator), as there is high probability that it will make User A feel better.

User A's reaction to the system's humorous utterance is also analyzed by the ML-Ask affect analyzer, and the output of this analysis is again stored in the humor sense model.

Similar process occurs when users joke. The system's utterance that precedes the joke is analyzed by the individualization module, and the data is stored in the user's humor sense model. For example, if the system's utterance is classified as containing the emotion of "frustration", and the user's reaction to it is recognized as humorous, the individualization module will store this information to use it in further interactions (<tells jokes when partner's emotional state is <frustration>).

This individualization module's algorithm allows the system to „learn" users' humor sense by gathering data during conversations. Such approach, however, has one serious drawback: in order to build such model, the system has to perform numerous conversations, in which it's performance will not be personalized. This means that there is a risk that some users may not be willing to continue the interactions after being told jokes that do not match their sense of humor.

In our previous works [25,26] we used a simple rule: if user's emotive state was negative or neutral, the system would use humor to make him/her feel better. This rule could also be used as an initial setting of the individualization module. However, in our opinion it is overly general and thus we are currently working on another, statistical approach.

Although humor is a highly individual issue, there are some regularities of its perception depending on receiver's age, sex, education background or social status. Japanese puns (*dajare*), for instance, are often called "old men jokes" (*oyaji gyagu*), which

suggests that they are liked the most by males above certain age. Dirty jokes may be preferred by male audience, while more subtle humor could be appreciated more by females. In order to investigate this issue, we are currently conducting a survey, in which we ask participants to assess examples of several types of humor.

The results will be used to set the initial settings of our system. Knowing that, for instance, middle-aged male company workers tend to like jokes on politics, the system will use this knowledge in initial humor sense model of every user representing this group. Such models will later be modified according to the data acquired for each particular user.

In our research we focus on the role of humor in conversations between the system and users. However, the proposed individualization module can be used also to gather other types of data, such as users' reactions to particular topics of conversation.

5 Conclusion and future work

In above sections we briefly presented current development stage of our project, aimed at creating a conversational system that uses humor in an individual manner, to make its human users feel better. This is still an ongoing work, and currently we are focusing on improving its particular modules. In the nearest future we are planning to conduct initial evaluation experiments and use their results in the system's further improvement.

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